Enabling Technological Progress and American AI Dominance through the Diffusion of AI Knowledge in the AI Action Plan

Comment of Professor Kevin Frazier on the Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

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I. Introduction

I respectfully submit this Comment to the National Science Foundation ("NSF") in response to its Request for Information on the Development of an Artificial Intelligence ("AI") Action Plan.[1] I am a Professor of Law at the St. Thomas University Benjamin L. Crump College of Law, a Fellow of the Lawfare Institute, and a Senior Research Affiliate of the Institute for Law and Artificial Intelligence; my expertise is in national security and AI, and I have published extensively on topics such as innovation, individual liberty, and American dominance in AI.[2] Thus, I welcome the opportunity to provide input on President Trump's upcoming AI Action Plan.

This Comment recommends that the AI Action Plan prioritize a national strategy for AI literacy—one that systematically breaks down geographic barriers to AI knowledge and creates pathways for all Americans to participate in the AI economy. Recent research reveals a troubling reality: communities more than 125 miles from AI hotspots see 17% lower growth in AI-related jobs and innovation, creating widening opportunity gaps between coastal tech hubs and the rest of America.[3] A national strategy for AI literacy would support American AI technological progress and American AI dominance through the diffusion of scientific knowledge and expertise.

For the U.S. to maintain technological leadership, it must do more than establish AI literacy in a few centers—it demands a comprehensive approach to diffusing AI knowledge across the country, much as the Rural Electrification Administration once transformed America by spreading both electrical infrastructure and practical knowledge to communities far from

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[1] National Science Foundation. "Request for Information on the Development of an Artificial Intelligence (AI) Action Plan." Federal Register (90 FR 9088), February 6, 2025. https://www.federalregister.gov/documents/2025/02/06/2025-02305/request-for-information-on-the-development-of-an-artificial-intelligence-ai-action-plan.

[2] Adjunct Professor at Delaware Law; Emerging Technology Scholar at St. Thomas University College of Law.

[3] Hunt, Jennifer, Iain M. Cockburn, and James Bessen. "Is Distance from Innovation a Barrier to the Adoption of Artificial Intelligence?" Working Paper. Working Paper Series. National Bureau of Economic Research, October 2024. https://doi.org/10.3386/w33022.

urban centers. The stakes of this challenge extend beyond economic metrics to the very foundations of American competitiveness in an AI-driven future.

My recommendations for the AI Action Plan are summarized below in Table 1.

Time Frame	**Recommendations**
Short-term (1–4 years)	● Creating a national AI extension service, modeled after agricultural extension programs, to bring AI expertise to underserved areas ● Establishing regional AI resource centers that can serve as intermediaries between major research hubs and local communities ● Developing standardized AI literacy curricula that can be rapidly deployed through existing educational institutions ● Incentivizing AI experts to spend time teaching and consulting in regions currently lacking AI expertise
Medium-term (5–10 years)	● Supporting the development of AI faculty at regional universities and community colleges ● Creating partnerships between AI hotspot institutions and more distant educational institutions ● Developing specialized AI programs tailored to regional economic needs ● Establishing AI research facilities in currently underserved regions to create new knowledge hubs
Long-term (10+ years)	● Building a national platform for AI continuing education

	● Creating mechanisms for sharing AI expertise between industries and regions ● Developing standards for AI education and certification ● Establishing frameworks for public-private partnerships in AI education

Table 1: Recommendations for President Trump's AI Action Plan

II. Current Gaps in the Diffusion of AI Knowledge and Expertise

The current distribution of AI expertise and understanding in the United States reveals concerning disparities. A growing body of research—including Professor Hunt's research—demonstrates that there is a stark geographic concentration of AI knowledge in pre-existing innovation hubs. The "superstar" cities of the last wave of technological innovation are doubling down on their talent and infrastructure to thrive in the AI era.[4] While their success is laudable, the fact that the rest of America is missing out raises concerns. "AI innovation hotspots," to borrow Hunt's term, have disproportionately high rates of AI hiring, growth in AI R&D, and overall economic growth. Those economic and knowledge gains, though, are not spreading. A lack of digital skills and knowledge persists in communities with lower educational attainment,[5] less economic security,[6] and more rural geographic settings.[7]

These geographic disparities in AI expertise manifest in several ways. Large enterprises, particularly those near AI innovation centers, are rapidly incorporating AI capabilities; Hunt's work shows that finance and insurance industries, traditionally concentrated in major metropolitan areas, are seeing some of the fastest growth in AI job postings. Meanwhile, businesses in sectors like retail trade and warehousing, often located away from tech hubs, show markedly slower AI adoption rates[8]—as do smaller businesses.[9] The differences in AI uptick

[4] Muro, Mark, and Yang You. "Superstars, Rising Stars, and the Rest: Pandemic Trends and Shifts in the Geography of Tech." Brookings Institute, March 8, 2022. https://www.brookings.edu/articles/superstars-rising-stars-and-the-rest-pandemic-trends-and-shifts-in-the-geography-of-tech/.

[5] Anderson, Emily A. Vogels and Monica. "Americans and Digital Knowledge in 2019." *Pew Research Center* (blog), October 9, 2019. https://www.pewresearch.org/internet/2019/10/09/americans-and-digital-knowledge/.

[6] everyoneon, and John B. Horrigan. "Digital Skills and Trust: How They Affect the Way Low- and Lower-Middle Income Households Connected to the Internet during the Pandemic." everyoneon, February 2022. https://static1.squarespace.com/static/5aa8af1fc3c16a54bcbb0415/t/61fc71248a56247e899c2a20/1643933997111/EveryoneOn_Report_2_Digital_Skills_and_Trust.pdf.

[7] Daepp, Madeleine I. G., and Scott Counts. "The Emerging AI Divide in the United States." arXiv, April 30, 2024. https://doi.org/10.48550/arXiv.2404.11988.

[8] Hunt et al., "Is Distance from Innovation a Barrier to the Adoption of Artificial Intelligence?"; McElheran, Kristina, J. Frank Li, Erik Brynjolfsson, Zachary Kroff, Emin Dinlersoz, Lucia Foster, and Nikolas Zolas. "AI Adoption in America: Who, What, and Where." *Journal of Economics & Management Strategy* 33, no. 2 (2024): 375–415. https://doi.org/10.1111/jems.12576.

[9] McElheran et al., "AI Adoption in America: Who, What, and Where."

among different industries means that while one region's workforce upskills on a daily basis, other regions will see their laborers struggle to keep up with new AI advances. Workers in the latter communities both now and for the foreseeable future will find themselves on the wrong side of the tech boom. As time passes, their counterparts in AI hotspots will accumulate ever more knowledge as to the risks and benefits of AI.

The education sector presents another troubling dimension of this knowledge gap. While universities in AI hotspots are rapidly expanding their AI curricula and research programs to cater to local employers, educational institutions outside of these areas lack the faculty expertise or resources to offer meaningful AI education.[10] Hunt et al.'s finding that distance from AI research centers significantly impacts the growth of AI-related educational opportunities suggests a self-reinforcing cycle of knowledge concentration.

Perhaps most concerning is the trend of leading AI researchers gravitating toward private industry rather than public-facing roles. Hunt et al.'s research notes that companies in AI hotspots can offer significantly higher compensation, drawing talent away from positions that might help spread AI knowledge more broadly. Industry organizations are also able to provide more access to computing power,[11] and the proportion of industry-affiliated research has only risen with the dominance of compute-demanding large language models.[12] This brain drain from public institutions further concentrates AI expertise in select private companies and locations.

III. Diffusing Emerging Technology Knowledge is Essential to Technological Progress

Historical precedents demonstrate that broad diffusion of technological knowledge is crucial for maximizing societal benefits.[13] The electrification of America offers an instructive parallel. While initial electrical innovations were concentrated in urban centers, the 20th Century's Rural Electrification Administration recognized that spreading both electrical infrastructure and knowledge about its applications was essential for national development. This dual focus on physical infrastructure and knowledge diffusion transformed American agriculture and rural industry.[14] By supporting local cooperatives and private utilities in expanding service to rural communities, the Administration helped double the number of rural farms with electricity within just five years.[15]

[10] Sun, Jeffrey C., and Taylor L. Pratt. "Navigating AI Integration in Career and Technical Education: Diffusion Challenges, Opportunities, and Decisions." *Education Sciences* 14, no. 12 (December 2024): 1285. https://doi.org/10.3390/educsci14121285.
[11] Besiroglu, Tamay, Sage Andrus Bergerson, Amelia Michael, Lennart Heim, Xueyun Luo, and Neil Thompson. "The Compute Divide in Machine Learning: A Threat to Academic Contribution and Scrutiny?" arXiv, January 8, 2024. https://doi.org/10.48550/arXiv.2401.02452.
[12] Abdalla, Mohamed, Jan Philip Wahle, Terry Ruas, Aurélie Névéol, Fanny Ducel, Saif Mohammad, and Karen Fort. "The Elephant in the Room: Analyzing the Presence of Big Tech in Natural Language Processing Research." In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, edited by Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, 13141–60. Toronto, Canada: Association for Computational Linguistics, 2023. https://doi.org/10.18653/v1/2023.acl-long.734.
[13] Ding, Jeffrey. *Technology and the Rise of Great Powers: How Diffusion Shapes Economic Competition*. Princeton University Press, 2024. https://www.jstor.org/stable/jj.12228596.
[14] Ronald R. Kline, Ph D. *Consumers in the Country*. Johns Hopkins University Press, 2000. https://doi.org/10.56021/9780801862489.
[15] Kitchens, Carl, and Price Fishback. "Flip the Switch: The Impact of the Rural Electrification Administration 1935–1940." *The Journal of Economic History* 75, no. 4 (December 2015): 1161–95. https://doi.org/10.1017/S0022050715001540.

The internet's development provides a more recent example. While early internet innovations emerged from concentrated research centers, deliberate efforts to spread both internet access and digital literacy were crucial for realizing the technology's full potential. The creation of the National Research and Education Network by President George H. W. Bush and subsequent educational initiatives helped ensure that internet knowledge wasn't confined to technical specialists.[16]

Hunt et al.'s research supports this historical pattern. They find that regions with greater AI knowledge diffusion, as measured by the presence of both research and practical applications, show faster overall AI adoption and innovation. This suggests that concentrating AI expertise in a few centers may actually slow down broader technological progress.

Moreover, broader technological understanding supports more informed public discourse and policy-making. When the public has a basic grasp of a technology's capabilities and limitations, policy discussions can focus on substantive issues rather than misconceptions. This is particularly crucial for AI, where public misconceptions could lead to either unwarranted fear or dangerous overconfidence.

IV. Diffusion of AI Knowledge is Essential to America's AI Dominance

The research suggests that America's continued AI leadership requires more than just concentrated excellence in a few centers—especially as allies and competitor nations are racing ahead on promoting the AI knowledge diffusion. China's AI strategy has long emphasized a need to "comprehensively improve the level of the whole society on the application of AI."[17] Just last year, Chinese Premier Li Qiang announced an "AI Plus" initiative as part of an effort to "fully integrate digital technology into the real economy,"[18] and AI is being incorporated at all levels of Chinese public education.[19] Similarly, Russian President Vladimir Putin has stated that "the most important issue [facing Russia] is whether our society and people are ready for the pervasive implementation of [AI]," calling for "mass digital literacy and re-training programmes."[20]

[16] McClure, Charles R. "Network Literacy: A Role for Libraries?" *Information Technology and Libraries* 13, no. 2 (1994): 115–25.
[17] State Council of the People's Republic of China. "Full Translation: China's 'New Generation Artificial Intelligence Development Plan' (2017)." Translated by Rogier Creemers, Graham Webster, Paul Triolo, and Elsa Kania, August 1, 2017. https://digichina.stanford.edu/work/full-translation-chinas-new-generation-artificial-intelligence-development-plan-2017/.
[18] Qiang, Li. "Report on the Work of the Government: Delivered at the Second Session of the 14th National People's Congress of the People's Republic of China on March 5, 2024." NPC Observer, 2024. https://npcobserver.com/wp-content/uploads/2024/03/2024-Government-Work-Report_EN.pdf.
[19] Ministry of Education of the People's Republic of China. "MOE Issues Guidance on How to Teach AI in Primary and Middle Schools - Ministry of Education of the People's Republic of China," December 2, 2024. http://en.moe.gov.cn/news/press_releases/202412/t20241210_1166454.html.
[20] Putin, Vladimir. "Meeting on the Development of Artificial Intelligence Technologies." President of Russia, May 31, 2019. http://en.kremlin.ru/events/president/news/60630.

Countries including France,[21] the UK,[22] Saudi Arabia,[23] Singapore,[24] Malaysia,[25] and Canada[26] have recognized the importance of or are actively promoting national AI literacy. Maintaining competitiveness in AI will require that the U.S. does the same.

As an aside, which of the aforementioned national efforts merits emulating is unclear. The UK spun its program up in early 2025. France's program started in 2018 and has already managed to significantly increase the number of computer science graduates. The CanCode program in Canada, established in 2017, has likewise documented a substantial uptick in AI knowledge among its hundreds of thousands of young participants.[27] A commitment to AI literacy in the US may involve borrowing parts of these and other programs.

Preventing uneven adoption rates is crucial for maintaining public support for AI development. If some regions are indeed falling significantly behind in AI adoption, then the country is at risk of creating "AI deserts" where lack of exposure breeds resistance to the technology. This could lead to political pushback against AI development, similar to historical reactions against other unevenly distributed technological changes.

More broadly distributed AI knowledge would also help address workforce adaptation challenges. Regions with better access to AI knowledge see higher growth in jobs adapting AI to new uses, rather than just jobs being replaced by AI. This suggests that spreading AI literacy could help more workers transition to AI-enhanced roles rather than being displaced.

V. Recommendations for the AI Action Plan on the Diffusion of AI Knowledge

Drawing on research findings[28] about geographic barriers to AI knowledge diffusion, the AI Action Plan should consider several time-horizons for spreading AI literacy:

Short-term initiatives (1-4 years) should focus on breaking down immediate barriers to AI knowledge diffusion. The research shows that physical distance from AI hotspots significantly impedes knowledge transfer. To counter this, the Plan could propose:

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[21] European Commission. "France AI Strategy Report," September 1, 2021. https://ai-watch.ec.europa.eu/countries/france/france-ai-strategy-report_en.
[22] UK Secretary of State Science, Innovation and Technology. "AI Opportunities Action Plan." Department for Science, Innovation & Technology, January 13, 2025. https://www.gov.uk/government/publications/ai-opportunities-action-plan/ai-opportunities-action-plan.
[23] Saudi AI & Data Authority. "Realizing Our Best Tomorrow: Strategy Narrative." National Strategy for Data & AI, October 2020. https://web.archive.org/web/20240620161900/https://ai.sa/Brochure_NSDAI_Summit%20version_EN.pdf.
[24] Wong, Laurence. "PM Lawrence Wong at the Launch of Smart Nation 2.0." Text. Prime Minister's Office Singapore. October 3, 2024. https://www.pmo.gov.sg/Newsroom/PM-Lawrence-Wong-at-the-Launch-of-Smart-Nation.
[25] Prime Minister's Office of Malaysia. "Govt Serious About Digital Transformation, Promoting AI Literacy - PM Anwar," January 16, 2024. https://www.pmo.gov.my/2024/01/govt-serious-about-digital-transformation-promoting-ai-literacy-pm-anwar/.
[26] Trudeau, Justin. "Statement by the Prime Minister on International Literacy Day." Prime Minister of Canada, September 7, 2024. https://www.pm.gc.ca/en/news/statements/2024/09/08/statement-prime-minister-international-literacy-day.
[27] Innovation, Science and Economic Development Canada. "Evaluation Report of CanCode." Government of Canada. Innovation, Science and Economic Development Canada, September 17, 2024. https://ised-isde.canada.ca/site/audits-evaluations/en/evaluation/evaluation-report-cancode.
[28] Hunt et al., "Is Distance from Innovation a Barrier to the Adoption of Artificial Intelligence?"

- Creating a national AI extension service, modeled after agricultural extension programs, to bring AI expertise to underserved areas
- Establishing regional AI resource centers that can serve as intermediaries between major research hubs and local communities
- Developing standardized AI literacy curricula that can be rapidly deployed through existing educational institutions
- Incentivizing AI experts to spend time teaching and consulting in regions currently lacking AI expertise

Medium-term strategies (5-10 years) should focus on building sustainable AI education capacity across the country. Hunt et al.'s finding that distance reduces growth in both research and practical AI applications suggests the need for developing local expertise. Key considerations include:

- Supporting the development of AI faculty at regional universities and community colleges
- Creating partnerships between AI hotspot institutions and more distant educational institutions
- Developing specialized AI programs tailored to regional economic needs
- Establishing AI research facilities in currently underserved regions to create new knowledge hubs

Long-term initiatives (10+ years) should aim to create an ecosystem for continuous AI learning and adaptation. The paper's evidence that state borders impede knowledge flow suggests the need for stronger national infrastructure for AI knowledge sharing. Components might include:

- Building a national platform for AI continuing education
- Creating mechanisms for sharing AI expertise between industries and regions
- Developing standards for AI education and certification
- Establishing frameworks for public-private partnerships in AI education

The Plan should distinguish between different levels of AI knowledge needed because different industries and occupations require varying levels of AI expertise. General AI literacy—basic understanding of AI capabilities, limitations, and implications—is needed broadly. More specialized knowledge—such as AI system development or domain-specific AI applications—is needed in specific sectors and roles.

VI. Conclusion

Geographic disparities in AI knowledge suggest that current approaches to AI development may be suboptimal. While concentrated centers of AI excellence have driven impressive advances, the barriers to knowledge diffusion are limiting broader technological progress and potentially creating societal divisions.

The AI Action Plan offers an opportunity to address these challenges by taking a more comprehensive approach to AI knowledge diffusion. By considering short-, medium-, and long-term initiatives to spread both general and specialized AI knowledge, the Plan could help ensure that America's AI leadership is built on a broader and more sustainable foundation than just a few centers of excellence.

Success in the global AI race will require both maintaining America's leading edge in advanced AI research and ensuring that AI knowledge and capabilities are widely distributed throughout the country. As Hunt et al.'s research demonstrates, distance from AI hotspots currently impedes this distribution. By deliberately addressing these barriers through a comprehensive approach to AI knowledge diffusion, the AI Action Plan could help ensure that America's AI leadership is both technologically advanced and societally sustainable.

This balanced approach—maintaining concentrated excellence while actively spreading knowledge—may be essential for achieving the Executive Order's goal of enhancing America's AI dominance. It would create a more resilient AI ecosystem less dependent on a few geographic centers and better able to adapt to unexpected technological developments like DeepSeek's recent advances.

The success of such an approach will require sustained commitment and coordination across government, industry, and academia. However, the potential benefits—from more widespread AI innovation to better-informed public discourse about AI—suggest that such investment in AI knowledge diffusion could be crucial for America's technological future.